

Branching & Merging Practice: Todo Application

You are joining a project as a developer.

You have been given an existing Todo application. Your job is to implement the product tickets below and manage your work using Git and GitHub.

Project Provided to You

The repository already contains a working Todo application.

Existing functionality

- Add a Todo
- Mark a Todo as completed
- Delete a Todo
- Display all Todos

Existing project structure

You don't need to create the project from scratch.

You will receive:

```
todo-app/  
├── src/  
│   ├── components/  
│   │   ├── TodoForm  
│   │   ├── TodoList  
│   │   └── TodoItem  
│   ├── data/  
│   │   └── todos  
│   └── App  
├── README.md  
├── package.json  
└── ...
```

Your Workflow

For **every ticket**:

1. Start from the latest `main`.
2. Create a new feature branch.
3. Implement the ticket.
4. Make meaningful commits.
5. Push the branch to GitHub.
6. Create a Pull Request.
7. Review your changes.
8. Merge the Pull Request into `main`.
9. Delete the feature branch after merging.

Do not directly develop features on `main`.

Use meaningful branch names such as:

feature/todo-search
feature/todo-priority
feature/todo-category

Ticket 01 — Todo Search

Requirement

Users currently have to manually scan the Todo list.

Add a search feature.

Requirements

- Add a search input above the Todo list.
- Search should filter Todos by title.
- Searching should be case-insensitive.
- The list should update as the user types.
- Display an appropriate message when no Todo matches.

Git

Create a branch:

feature/todo-search

Complete the feature and merge it into main.

Ticket 02 — Todo Priority

Requirement

Users need to identify important Todos.

Add priority support.

Requirements

Every Todo should have one of:

Low
Medium
High

Update the Todo creation form so the user can select a priority.

Display the priority on each Todo.

Use different visual styles for each priority.

Example:

High → Red
Medium → Orange

Low → Gray

Existing Todos should continue to work.

Git

Create a branch:

feature/todo-priority

Complete the feature and merge it into main.

Ticket 03 — Todo Card Redesign

Requirement

The product team wants the Todo cards redesigned.

Update each Todo card so that:

- The Todo title is displayed prominently.
- Priority is displayed beside the title.
- The complete button appears on the right.
- Delete remains available.
- Completed Todos have a different visual appearance.
- High-priority Todos have a clear visual indicator.

Keep the existing functionality working.

Git

Create:

feature/todo-card-redesign

Complete the ticket and create a Pull Request.

Do not merge it yet.

Ticket 04 — Todo Categories

Requirement

Users want to organize Todos into categories.

Add the following categories:

Work
Personal
Other

Requirements

- Add category selection to the Todo form.
- Store the category with each Todo.

- Display the category on the Todo card.
- Add a category filter.
- Existing Todos should continue to work.

Git

Create:

feature/todo-categories

Complete the feature and create a Pull Request.

Now merge the open feature branches into `main`.

If Git reports a conflict, resolve it while making sure **both product requirements continue to work**.

After resolving the conflict, test:

- Todo priority
- Todo card design
- Todo categories
- Existing Todo functionality

Ticket 05 — Todo Editing

Requirement

Users currently cannot correct a Todo after creating it.

Add an Edit feature.

Requirements

Users should be able to edit:

- Title
- Priority
- Category

The existing Todo should be updated rather than creating a duplicate.

Canceling an edit should leave the Todo unchanged.

Git

Create:

feature/edit-todo

Implement the feature and merge it through a Pull Request.

Ticket 06 — Todo Statistics

Requirement

The product team wants a quick overview of the Todo list.

Add a statistics section above the Todo list.

Display:

Total
Active
Completed
High Priority

The numbers must update automatically when Todos are added, deleted, completed, or edited.

Git

Create:

`feature/todo-statistics`

Implement and merge through a Pull Request.

Ticket 07 — Todo Activity

Requirement

Users want to know what happened to their Todos.

Add an activity/history section.

For each Todo, record relevant actions such as:

Todo created
Todo edited
Priority changed
Category changed
Todo completed
Todo deleted

Display the activity in the Todo details view.

Git

Create:

`feature/todo-activity`

Implement the feature.

Before merging your branch, update your local branch with the latest `main`.

Resolve any conflicts that occur.

Then push the branch and create the Pull Request.

Ticket 08 — Final Product Update

The product team has requested one final improvement.

Add:

Dark Mode

Users should be able to switch between:

Light Mode
Dark Mode

The selected mode should apply to the entire application.

Make sure existing functionality continues to work.

Git

Create:

feature/dark-mode

Implement the feature and merge it through a Pull Request.

Final Requirements

At the end of the exercise:

- All tickets should be implemented.
- All features should be merged into `main`.
- Every feature should have been developed on a separate branch.
- Every feature should have a Pull Request.
- Commits should have meaningful messages.
- The application should work after all merges.
- Your GitHub repository should contain the complete development history.

Most importantly, you should be able to look at your Git history and understand:

- Which feature was developed in which branch.
- Which changes were merged into `main`.
- Why a merge conflict happened.
- How you resolved the conflict.
- What you did when your branch was behind `main`.
- Why you used rebase.